A PROJECT TITLE

ON

Multi-Class Disease Diagnosis Using Multinomial Logistic Regression

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF " Computer Science and Information Technology"

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' Multi-Class Disease Diagnosis Using Multinomial Logistic Regression'

1. Abstract:

Healthcare systems generate vast amounts of patient data, including demographic details, clinical symptoms, and laboratory test results. Traditional diagnostic methods often rely on manual interpretation, which can be slow and prone to error. To address these challenges, this project applies **Multinomial Logistic Regression (MLR)** to the problem of multi-class disease diagnosis. Unlike binary logistic regression, which predicts only two outcomes, MLR extends the model to classify patients into multiple disease categories simultaneously, making it highly relevant for real-world medical applications.

The proposed system builds a predictive framework that models the probability distribution across several disease classes. This allows healthcare professionals to identify the most likely diagnosis while also considering alternative possibilities. Such probabilistic outputs enhance interpretability, enabling practitioners to evaluate risk factors and prioritize further testing or treatment. The model' s transparency is a key advantage, as it provides insight into how individual features influence predictions.The workflow includes data preprocessing, feature selection, model training, and evaluation. Performance is assessed using metrics such as accuracy, precision, recall, and F1-score. Visualization techniques are employed to highlight the role of different features in disease prediction, ensuring clarity and trust in the results. The system is designed to be scalable, supporting integration with larger datasets and potential deployment in clinical decision support platforms.

By combining the robustness of logistic regression with the flexibility of multi-class classification, this project demonstrates how machine learning can improve diagnostic accuracy, reduce misclassification, and support early intervention. Ultimately, the model provides a practical and interpretable solution for automated disease diagnosis, contributing to the broader goal of leveraging data-driven methods to enhance healthcare outcomes and optimize patient care.

2. INTRODUCTION:

Healthcare systems generate vast amounts of patient data, ranging from clinical symptoms to laboratory test results. Traditional diagnostic methods often rely on manual interpretation, which can be time-consuming and prone to human error. With the rise of **Machine Learning (ML)**, automated disease diagnosis has become a promising solution.

Logistic Regression is a widely used statistical model for classification tasks. However, standard logistic regression is limited to binary outcomes. In real-world medical scenarios, patients may present with multiple possible diseases, requiring a **multi-class classification approach**. Multinomial Logistic Regression addresses this challenge by modeling the probability of each disease category relative to a baseline class.

This project applies MLR to disease diagnosis, aiming to:

- Automate classification of patient records into multiple disease categories.
- Improve diagnostic accuracy using statistical learning.
- Provide a scalable solution for healthcare data analysis.

The system is designed to be interpretable, allowing medical practitioners to understand the influence of each feature on disease prediction, thereby ensuring transparency in clinical decision-making.

3. SOFTWARE REQUIREMENTS:

- **Programming Language**: Python
- **Libraries**: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn
- **ML Frameworks**: Scikit-learn (for Logistic Regression), TensorFlow or PyTorch (optional for extensions)
- **Data Handling**: CSV/Excel files for patient records, SQL/NoSQL databases for larger datasets
- **Development Environment**: Jupyter Notebook, PyCharm, or Visual Studio Code
- **Version Control**: Git and GitHub
- **Operating System**: Windows, Linux, or macOS
- **Visualization Tools**: Matplotlib, Seaborn, Plotly (for interactive plots)

Signature of the Guide

Course Instructor